

Resorbable Collagen Barrier Impeding the Extrusion of Obturating Material in Primary Molars Undergoing Resorption – A Randomized Clinical Trial

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INTRODUCTION

The major goal of a pediatric dentist is to maintain the integrity of primary dentition until normal exfoliation.¹ The assessment of accurate morphology of root canal system determines affluent root canal management in deciduous molars. The physiologic resorption of primary teeth, which initiates shortly after the root development, modifies the anatomy of root canal.^{2,3} Also, pathological process due to pulp and/or periodontal inflammation hastens the root resorption.^{4,5} The most viable treatment option to avoid premature loss of deciduous teeth is pulpectomy. Retention of primary dentition aids to maintain the arch length, preserve occlusal function and provide eruption guidance for the permanent teeth.⁶

In primary teeth undergoing resorption, complete debridement of canal space and achieving an apical seal is an arduous task. Despite of myriads of obturation materials and techniques used, overfilling/over extrusion of obturating material is a common outcome in the deciduous teeth which may pave way to deleterious effects on the underlying permanent dental follicle. Application of resorbable apical biological barrier in the form of collagen is a well-practiced strategy for apexification procedure.⁷ Also it has been widely used as a pulp dressing material following pulpotomy in primary teeth.⁸ These materials act as an interface between restorative materials and biological tissues. During the evolving year's, much progress has been made in expanding the scope of collagen in soft tissue engineering and repair.⁹⁻¹²

Johnson *et al*¹³ did an *in vitro* study to prevent extrusion of obturating material beyond apex by placing resorbable collagen barrier at apical one third of the root canal and found that not completely but it significantly plummeted the possibility for over obturation in resorbing primary molars. To the best of our knowledge, the literature lacks sufficient evidence regarding the efficacy of placing an apical barrier in resorbing primary teeth. This randomized clinical trial was undertaken to find out whether positioning of a resorbable barrier of collagen have any significant effect in impeding the extrusion of obturating material in primary molars undergoing resorption.

MATERIALS AND METHOD

The institutional ethics committee received the study protocol and gave its approval vide Ref. No: TMDCRC/IEC/18-19/PPD1 dated 22/1/2019. A total of 60 child patients were screened at Department of Pedodontics and Preventive Dentistry, and 47 children who gratified the inclusion criteria demanding pulpectomy procedure in 47 primary mandibular molar teeth were included in this study. It was a randomized clinical trial in which 94 canals in 47 primary molars were equally allocated to Group A and Group B, based on randomization protocol and all the procedures were performed and evaluated by a single operator (N.M.S). Calculation of sample size was done following power analysis which was 80% for this study. After parent's consent, pulpally involved mandibular primary molars with at least two third of root length in children aged 7-9 yrs¹⁴ with behavior rating of Frankel's Class 3 or 4 without any systemic disease were included in the study.^{15,16} However, perforation of pulpal floor due to resorption, extreme tooth mobility, dentition with congenitally missing permanent tooth bud, radiographic sign of any internal or external resorption and radiolucencies in radicular area were excluded from the study.

One mesial and one distal canal of each selected tooth were randomly assigned to test or control group by using flip coin method by the parent. After that local anesthesia was administered, rubber dam isolation was done followed by entry into the pulp chamber by means of a high speed sterile round bur sequenced by pulp extirpation and copious irrigation with 3% sodium hypochlorite and normal saline. The location and number of root canals were recorded using a double ended endodontic explorer. After pulp extirpation, each canal's working length was radiographically determined by using No. 15 k file of 21 mm length. In order to accommodate 2mm of apical barrier, the final instrumented length of test canal was kept 2mm minus radiographic apex. To maintain standardization, the final working length of control canal was also kept 2mm short of radiographic apex.¹³ After thorough debridement and canal preparation with hand protaper files (SX till F2) the canals were dried to receive obturation.

A silicone stop cone was placed on an endodontic plugger and set to final working length. A 2mm square piece of resorbable collagen (Coloplug, Cologenesis Healthcare Pvt Ltd, Tamilnadu, India) (Figure-1) was placed into the test canal and condensed till the final instrumented length (2mm short of the radiographic apex) using the same endodontic plugger. The selected canals (test and control) were then obturated by a pre-packed syringe delivery system containing a mixture of calcium hydroxide and iodoform paste (Metapex, Meta Biomed Co Ltd, Chuncheongbuk-Do, South Korea). The tip was positioned into the canal and the paste was

pushed gradually withdrawing the syringe till the obturation material streamed back into the pulp chamber. With the help of a moist cotton pellet, material was lightly pushed into the canal. All the remaining canals were obturated in the similar way.

After obturation of all the canals, a postoperative radiograph was obtained using radiovisiography (RVG)(Kodak Carestream RVG 5200, Health, Inc, NY, USA) (Figure- 2). Additional radiograph was done only in cases of superimposition. This was followed by glass ionomer cement restoration and placement of a stainless steel crown. The pre and post-operative digital radiographs for the presence of overfilling of the obturation material were assessed by the same operator. The canal showing optimal filling or the canal with 1 to 2 mm short of apex was considered as "acceptable" i.e. No Extrusion (NE). In case of extravasation of material into the periapex, it was considered as over filled i.e. Extruded (E) beyond apex.



without any pre-operative root resorption.

CONCLUSIONS

In the present study, extrusion of obturation material beyond apex was 6.4% in test canals while 17% in control canals. Although the comparison was statistically not significant, placing a resorbable barrier proved to be more successful in preventing extrusion of obturating material in resorbing primary molars. Future studies can involve advance radiological techniques like computed tomography scan, cone-beam computed tomography etc. for the exact evaluation of the extent and level of resorption. Also, other radiopaque resorbable materials may be used as apical barrier to prevent extrusion of obturating material. The concept of placing a resorbable barrier at apical third of canals should be considered while performing pulpectomies in resorbing primary molars.

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